

SABANCI UNIVERSITY

CS404-Artificial Intelligence

Fall 2025

Course Description	This course is a broad technical introduction to fundamental concepts and techniques in artificial intelligence. Topics include problem solving, search, knowledge representation and reasoning, reasoning and decision making under uncertainty. Other important topics and current application areas of artificial intelligence, such as automated planning, machine learning, computer vision, robotics, natural language understanding, and intelligent agents, will be discussed.
Instructor	Berrin Yanikoglu
Zoom link	https://sabanciuniv.zoom.us/j/93366735827 for occasional online lectures or office hours
Lecture Hours	Tuesdays 11:40-12:30 (FMAN G071) and Fridays: 3:40-5:30 (FMAN G071)
Office Hour	Fridays 10-11
Textbook	Textbook: Artificial Intelligence: A Modern Approach (AIMA), by Stuart Russell and Peter Norvig., 4th edition. It is a comprehensive book that widely used in the world for AI courses. Previous editions are simpler and adequate, but it is your responsibility to check the (small) differences with the final edition, if you happen to have those books passed on to you. Link to the book's web page, where you can find some useful information: http://aima.cs.berkeley.edu/
Time Conflicts & Attendance	- I generally introduce a new topic on the 1st hour of the week, so I will NOT give time conflict permission on that hour. - Similarly, I will NOT give time permission on the first hour of the 2-hour lectures. - I am assuming that you can cover the details of a topic yourselves, but you will be fully responsible if you sign up with time conflicts, including points lost on missed quizzes. - You will get your attendance point based on your presence/participation to pop quizzes.
Tentative Grading	Midterm (30%) & Final (35%) – in class Homeworks: 25% – 5 total (no programming, except one Python programming homework) Quizzes: 10% - announced ahead of time, in-class on Sucourse Attendance: Will only be used for official records. Passing Grade: - To pass the course you grade as calculated above must be at least 35 (strict) and "final exam grade" should be above 29/100 (you shd. be able to answer "near" one third of the question to claim to have learned the material). - If you take the final but get a failing grade (<30), you need to take the makeup as well, in which case the average of the final and the makeup will be considered as your final exam grade. You will fail the course if that weighted average is below 30. - Attendance & class participation will be taken into account in border-line grades. So make sure to attend lectures if you <i>have to pass the course</i>.
Exam Policies	- There will be one midterm, one final and one make-up (after the final). - The final exam will be in the final exam period (the University is scheduling exams but date not yet known). Who can take the Makeup Exam: Only people who miss the midterm or the final or who did not get a passing grade in the final can take the makeup. In the second case, the average of final and makeup will be counted as your final grade; otherwise, makeup will substitute the missing grade. Those who miss the exams will need to have a documented excuse (medical report etc) and advance warning.

Quizzes	• There will be one scheduled quiz every week, starting with the second week (3-5 questions, 3-5 minutes, 10pts), from the past material (typically just last week's). • Quiz will be at the end of the one-hour lecture on Tuesdays, in the last 5 minutes of the lectures, but occasionally on Fridays if Wednesday is a holiday etc (announced ahead of time), • There will be 12 scheduled quizzes total, and the lowest 2 quizzes will be ignored when computing the quiz grade. People who add the course during the Add/Drop period will have those quizzes not counted. • As quizzes are given a significant weight, the quiz grades will be capped at exam-average*1.5 (so if exam grade average is 60, your quiz grade will be capped at 90 even if your quiz average shows 100 points) to obtain a fairer grading.
Homeworks	There will be 5 homeworks total, with no separate project. Late homeworks will be accepted for 2 days with -10pt every late day (late submissions in the first hour will receive -3 to -5pts). Questions about homeworks should be asked at SUCourse Forum, and NOT on email, so that the answer helps everyone else who may have the same question. Students are also encouraged to answer questions, as long as the given answer does not include answers to the homeworks (e.g. you can talk about what was told in class, or what the homework asks, ..., but not how to do the homework).
LLM Usage (ChatGPT, Gemini etc)	You may use ChatGPT or similar tools for getting some programming help in your homeworks, but you are strongly suggested not to resort to their use in “thinking” or “knowledge” questions as questions similar to those will be replicated in exams. If used, you must mention LLM usage and extent (where/how). Also, any LLM usage must be to get some “help”, but not full answers. In fact, homeworks where the code/report were in mismatch possibly due to LLM usage, were given zero. Finally, you must NOT use LLMs in your quizzes or exams. It will be against academic honesty to use them, and disciplinary action will be taken if detected.
Syllabus	We will cover the chapters that are in blue, which constitute the introductory part of all the main components of AI. For these chapters, we will follow the book pretty closely (some small sections may be skipped) and will go in order. Students may take Computer Vision, Speech Understanding and NLP courses for the uncovered Part VI and Data and Artificial Intelligence Ethics course for part VII. Book web page: https://aima.cs.berkeley.edu/ Artificial Intelligence: A Modern Approach (AIMA) Russell and Norvig I Artificial Intelligence 1 Introduction ... 1 2 Intelligent Agents ... 36 II Problem-solving 3 Solving Problems by Searching ... 63 4 Search in Complex Environments ... 110 5 Adversarial Search and Games ... 146 6 Constraint Satisfaction Problems ... 180 III Knowledge, reasoning, and planning 7 Logical Agents ... 208 8 First-Order Logic ... 251 9 Inference in First-Order Logic ... 280 10 Knowledge Representation ... 314 11 Automated Planning ... 344 IV Uncertain knowledge and reasoning 12 Quantifying Uncertainty ... 385 13 Probabilistic Reasoning ... 412 14 Probabilistic Reasoning over Time ... 461 15 Probabilistic Programming ... 500 16 Making Simple Decisions ... 528 17 Making Complex Decisions ... 562 18 Multiagent Decision Making ... 599 V Machine Learning 19 Learning from Examples ... 651 20 Learning Probabilistic Models ... 721 21 Deep Learning ... 750 22 Reinforcement Learning ... 789 VI Communicating, perceiving, and acting 23 Natural Language Processing ... 823 24 Deep Learning for NLP ... 856 25 Computer Vision ... 881 26 Robotics ... 925 VII Conclusions 27 Philosophy, Ethics, and Safety of AI ... 981 28 The Future of AI ... 1012